



Protocol for MD simulations

iGEM KU Leuven 2023

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Guidance: prof. Mathy Froeyen

1. Introduction

This protocol was written with the guidance of prof. Mathy Froeyen from KU Leuven. All of the used scripts and the access to the resources were also provided by prof. Froeyen.

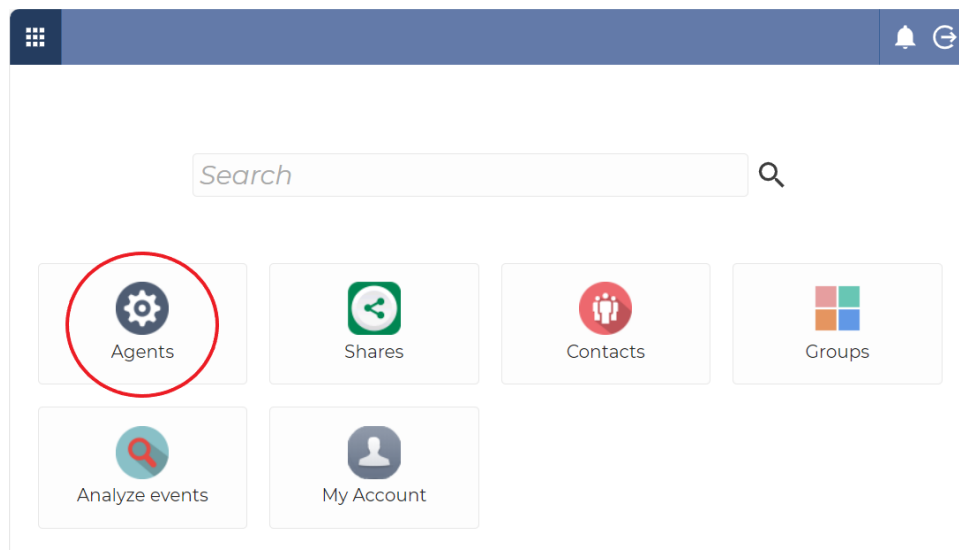
For the purpose of our project, we are performing Molecular Dynamics (MD) simulations of DHCR7 enzymes in complex with a docked ligand. In our case, the ligands are ergosta-5.7-dienol (since this is the substrate we're interested in) and NADPH which is needed for reducing ergosta-5.7-dienol to campesterol.

For MD simulations we're using Amber22. More information on this software can be found on [the Amber website](#). On the lab's computer, the Amber software package is installed in the `/usr/local/chem/amber/amber22/` directory. The Amber software is run by non-graphical script.

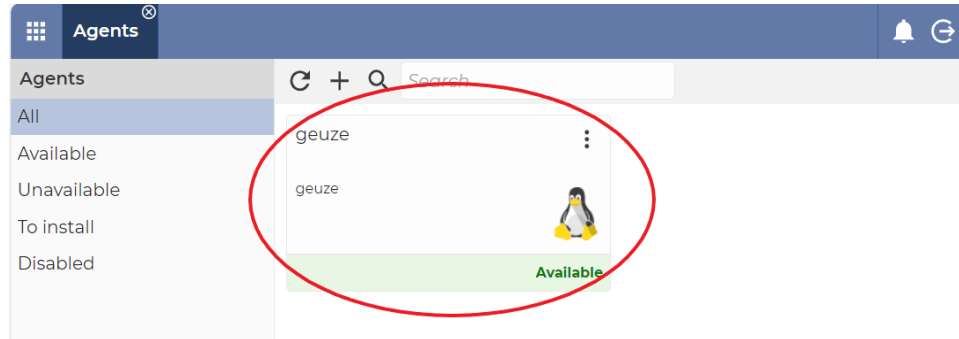
2. Remote access to the lab's computer

To make our lives easier, we've set up a remote connection to the computer in prof. Froeyen's lab with the use of [DWService](#). The connection through `ssh` was not possible at the moment.

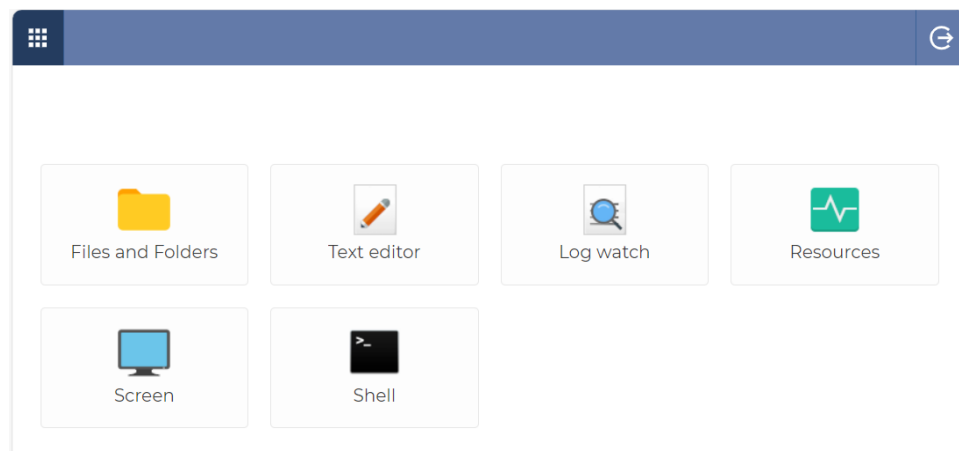
After logging in to the DWService account, click on the `Agents`.



Next, choose the `geuze` agent.



Now you have access to the files on the computer (`home/igem`) and you can use the terminal (no GUI, only command line).



3. Ligand preparation

This part is performed only once for each ligand. Since the force field for small molecules is not available, we need to prepare it ourselves. The result of this part of the protocol will act as a “reference” for our ligand.

Note: If you’re simulating a complex with more than one ligand, repeat those steps for each ligand (in separate folders) and specify all ligands in the `Run.leap.sol` script.



3.1. Retrieving the ligand structure

First, retrieve the 3D structure of the ligand from a database. In PubChem, the structure is not available in the PDB format, but in SDF format (among some others).

Download the SDF file of the ligand, next convert the format to PDB format using script `Run.sdf2pdb`. The script will produce a PDB file of the ligand.

```
./Run.sdf2pdb [ligand_filename]
```

Note: Do not specify the `.sdf` format in the `ligand_filename`.

Next, we need to change the atom names to unique ones. This can be done with the use of script `Run.restore_pdb`. The input file is specified as a first argument, the output as second. The output file name should have the `.out.pdb` ending (to distinguish it from the original version).

```
./Run.restore_pdb [ligand_filename.pdb] [ligand_filename.out.pdb]
```

3.2. Generating the force field parameters

With the use of programs `antechamber` and `parmchk2` available in the Amber package, we can generate force field parameters for our ligand. It is necessary to specify the desired net charge (in our case 0). For generating Amber force field parameters, we used the general AMBER force field 2 (GAFF2).

```
./Run.antechamber2 [ligand_filename (without .pdb)] 0  
./Run.parmchk2 [ligand_filename (without .pdb)] 0
```

`Antechamber` will produce a `ligand_filename.prep` file. It will be used by `parmchk2` that will create a `ligand_filename.frcmod` file. The `prep` file contains the topology (bonds, atoms), atom types and charges of the ligand. The `frcmod` file contains additional force field parameters not present in the general GAFF force field file.



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4. Complex preparation

Even though a force field for proteins is available, we still need to prepare the complex PDB file that we'll be using.

4.1. Manual prep

Since the PDB file of the docked complex was saved by us using Chimera, we need to make few manual modifications to the file. First, copy the file and give it a new name (ending with `.out.pdb`).

```
cp [complex_filename.pdb] [complex_filename.out.pdb]
```

Open the new PDB file in a text editor and remove all lines starting with `MODEL` and `ENDMDL`. Additionally, remove the lines mentioning the secondary structure (helix, sheet) from the beginning of the file. Do not leave any blank lines.

The numbering of the atoms in the ligand can differ between the docked ligand and the prepped ligand. Both ligand structures need to be investigated and matched (atom names + residue name). Take a look at lines describing the ligand in the ligand from the complex and the prepped ligand. The atom names (3rd column) and the "residue" name (4th column) of the docked ligand needs to match the prepped one. Change the atom names in the docked ligand to match the ones in the prepped ligand (**Important**: check everything in Chimera manually, atom by atom). You don't need to worry about missing hydrogens, Amber will take care of it later.

Normally you'll have to change the name of each atom to match it to the atom names in the reference ligands (`ligand_filename.out.pdb`). Remember to check the atoms manually in Chimera (open the complex file and the prepped ligand in separate Chimera windows and hover over the atoms). This step can be very frustrating and troublesome but it's necessary for the heavy atoms. You can delete the hydrogen lines from the ligand to make your life easier (later on `LEaP` will add the hydrogens by itself).

To change the name of the ligand "residue" you can use this command. Change `UNK` to any name that the ligand has in the 4th column of the complex PDB file.

```
sed -i -- 's/UNK/LIG/g' [complex_filename.out.pdb]
```

Structures of the enzyme in some of the organisms that we're analyzing have a long disordered C-terminal "tail". Because the structure of our enzyme is an AlphaFold model, we know that prediction of this region has low confidence, which means that we should not consider it as accurate. In addition, we know that



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this region is not functional, so we don't need to have it in our model. Open the complex PDB file in a visualization tool (like Chimera, PyMOL, VMD, etc.) and check the residues forming the "tail". Remove the lines of the residues that you want to delete from the PDB file using your preferred editor. Leave only a short fragment of the "tail".

This kind of disordered "tail" would result in creating a very big periodic box with a lot of water molecules which would lead to using more computational resources. Since we know that this region is not functional, we can delete it which will make our simulations less computationally demanding.

Note: When you delete residues from the C-terminal, there can be a difference in the amount of hydrogens in the new "first" residue. You might need to delete some hydrogen from that residue if there will be any errors in next steps.

Important: In next steps, the residues will be renumbered to start from 1. This will change the residue numbering so remember how many residues you deleted. This way you can calculate the "original" residue number and compare it with the original complex file.

4.2. LEaP

The LEaP program generates an Amber topology file and coordinate file for a structure given a particular force field. The needed LEaP commands are embedded in the `Run.leap.sol` script. In order for the script to work, lines 18 and 19 have to be filled with correct ligand files and line 21 has to contain the correct complex PDB file (in our case `[complex_filename.out.pdb]`).

```
./Run.leap.sol
```

Note: The LEaP program can complain about some protonated histidines -> open PDB in Chimera, select hydrogens (HD1/HE1) in histidines and delete them. Save the file to PDB and delete the lines mentioning the secondary structure. In fact, LEaP will add the missing hydrogens to the histidines with a correct name.

Chimera command to select atoms in particular residues (here: HD1 hydrogens in HIS residues):

```
select :HIS@HD1
```



5. MD simulation

5.1. Minimization

First, we perform energy minimization in order to relax the structure. The minimization is run with the use of `Run.min.cuda` script with default parameters on the GPU. This computer is equipped with a recent NVIDIA RTX3080ti graphics processing unit to speed up molecular dynamics calculations. Open the script using a text editor or in the terminal and change the number of residues in the restraint mask. This restraint mask is added to limit the movements of the enzyme/ligand while the solvent and ions are free to optimize their positions.

```
./Run.min.cuda
```

5.2. Heating

Next, we slowly heat up the system to the desired temperature. Our system is heated up to 300K (+/- 27 degrees Kelvin room temperature) with a 2fs timestep. Same as in the minimization script, remember to change the number of residues in the restraints mask the `Run.heat.cuda` script.

```
./Run.heat.cuda
```

5.3. Density

To adjust the density of the solvent, run the `Run.density.cuda` script with updated number of residues in the restraints mask.

```
./Run.density.cuda
```

5.4. Equilibration

The final step before production is equilibration. We perform our equilibration with 2500000 steps and 2fs timestep which calculates to a 5000ps (5ns) equilibration run. Since there are no restraints here, there is no mask, hence there is no need to adjust the residue number in the script.

```
./Run.equil.cuda
```



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After running the equilibrium, you can check the system by converting the restart file `equil.rst` file to a PDB file and opening it in a visualization software tool (Chimera, PyMOL, etc.) and see if nothing unfolded or if anything weird happened.

```
./Run.makepdb equil
```

After running the script `Run.makepdb`, the file `equil.pdb` is created, it can be opened and analyzed in any visualization tool (just like any other PDB file). Remember that the residue numbering in this file follows the internal Amber numbering.

Note: Conversion of a `.rst` file into a `.pdb` file can be done at any point during the analysis, not just after equilibration (and it should be done after each production step).

5.5. Production

Production is the longest part of the MD simulation. It's best to split production phase into few stages with fewer steps instead of running the whole production at once. This way it's easier to monitor the system and if the job crashes at some point, there is no need of rerunning everything.

Each stage of our production phase includes 2 million steps (with 2fs timestep) which sums up to 4ns. The number of production stages depends on the desired length of the simulation and the stability of the system. In our case, we perform 5 production stages which translates to 20-ns MD simulation (unless the system did not stabilize).

```
./Run.prod1.cuda  
./Run.prod2.cuda  
./Run.prod3.cuda  
./Run.prod4.cuda  
./Run.prod5.cuda
```

Important: Production stages are being run sequentially; it is vital that the previous stages finish before running the next one.

Every production run will produce some output files:

- `prod1.nc`: file containing intermediate structure coordinates of this production run
- `prod1.rst`: restart file; this one will be used by the next `Run.prod2.cuda` to start. This file will be continuously updated while `Run.md1.cuda` is running



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- `prod1.out`: log file of this run
- `prod1.info`: information file of current run while running; here you find information how long the current run will take.

Note: It is vital to monitor the RMSD after each production stage to ensure the stability of the system. Constant increase in the RMSD throughout the simulation suggests that the simulation time should be increased. If after the third production run the RMSD is still visibly rising, continue with more production stages until you see the RMSD stabilizing. When creating scripts for further production stages, you can copy the last stage's script and adjust the file names (numbering in particular) at the end of the script.

6. Analysis

6.1. RMSD

RMSD values can be analyzed using program `cpptraj`. Script `Run.rms2` calculates the RMSD values throughout the whole loaded trajectory. It is important to monitor RMSD after each production state and it's helpful to analyze the values throughout the whole simulation.

Open the `Run.rms2` script and include the trajectories that you're interested in (lines starting with `trajin`). Depending on what you're analyzing, you can either load one or multiple trajectory files. Adjust the name of the output file to the trajectory that you're analyzing. Remember to change the number of residues in the mask.

Perform analysis of RMSD values after each stage of production and include all preceding trajectories (after 1st production stage include only `prod1.nc`, after 2nd stage include `prod1.nc`, `prod2.nc`, after 3rd stage include all three trajectories: `prod1.nc`, `prod2.nc` & `prod3.nc`, etc.)

The values can be plotted for example with `xmgrace` or in some other way (e.g. with a python script or in Excel).

```
./Run.rms2  
  
xmgrace [rmsd_file]
```

Note: You will not be able to use `xmgrace` in the DWSERVICE terminal since there is no access to GUI this way. If you want to use `xmgrace`, you'll need to use it locally. You can also use a different approach (python script, Excel, etc.). You can use the `gracebat` command written below.



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```
gracebat [input.agr] -hdevice PNG -autoscale xy -printfile [output.png]
```

6.2. RMSF

RMSF values can also be calculated with `cpptraj`. These values can give us an indication on the fluctuation of the molecule. Using the `Run.tfactors2` script, you can calculate RMSF values for all backbone atoms.

```
./Run.tfactors2  
xmgrace [rmsf_file]
```

Run these commands after finishing the MD simulation. Make sure to load all trajectories and adjust for correct number of residues. Instead of `xmgrace` you can run `gracebat`.

```
gracebat [input.bfactor] -hdevice PNG -autoscale xy -printfile [output.png]
```

6.3. Hydrogen bonds

Since we're analyzing a complex of a receptor and a ligand, it's very important to keep the interactions between them in mind. In particular, hydrogen bonds contribute to specific binding and anchoring the ligand in the binding site. Therefore, we need to monitor the distance between the atoms that were forming hydrogen bonds (according to PoseView, LigPlot and/or visual inspection in UCSF Chimera).

The atoms forming the hydrogen bonds need to be adjusted in the script. You can also adjust the output file name in each `distance` command.

```
./Run.hbond_dist  
xmgrace [distance_file]
```

Run `gracebat` instead of `xmgrace` if you're working without a graphical interface.

```
gracebat [input.dat] -hdevice PNG -autoscale xy -printfile [output.png]
```



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The y-axis shows the distance between the selected atoms that are supposed to form a hydrogen bond. Usually, the cutoff for the distance between (heavy) atoms forming a hydrogen bond is around 3.5Å.

If the distance gets way bigger and doesn't come back to the usual hydrogen bond values, it can suggest that the hydrogen bond breaks. It is possible that a new hydrogen will be formed (with another residue). In that case, it's a good idea to check the neighboring residues and analyze distance between these atoms. You can plot multiple files using `xmgrace` if you list all of the desired files in the command.

If the distance between the analyzed atoms stays between the preferred distance for a hydrogen bond (2.7 - 3.5Å), it is most likely that the bond doesn't break during the simulation and the ligand is securely anchored in the binding site.

Important: If you deleted the C-terminal "tail" in your enzyme, remember that the residue numbering changed.

Important: It is always important, next to the `cpptraj` analysis, to check the structures from the MD runs visually in a graphical viewer like UCSF Chimera!

6.4. Conclusions

When drawing conclusions after the MD simulations and the analysis, we want to look out for anything suspicious or strange. Extremely high RMSD or RMSF values can suggest that something is wrong. The distance between the atoms forming a hydrogen bond can give an indication on how good the molecule is anchored in the binding site. It is also possible that the small molecule falls out of the binding pocket, which is not a good sign. We want to see stable RMSD, not many extreme RMSF spikes (but it's also possible) and hydrogen bonds formed between the ligand and receptor's residues.